

Take home test - Senior Clinical Data Scientist - Healthcare

This test is an opportunity to demonstrate your healthcare analytics expertise, product thinking, rapid prototyping skills, ability to communicate clearly with health system stakeholders, and judgment about how to turn EHR data into useful operational and quality-improvement insights.

Prior to your interview, please upload via Greenhouse the following (more details below):

1. Your dashboard code and any supporting analysis code.
2. A brief README with instructions to run the dashboard locally, including any packages, assumptions, and external public datasets used.
3. A short summary of your dashboard story, key insights, and recommended next actions for a health system administrator (2 paragraphs or less), including no more than 2 embedded figures or tables if helpful.
4. A short description of what would be needed to productize your dashboard or analytic approach. Please include key limitations of the current data, metrics or use cases that would be valuable for a health system but cannot be fully supported with the provided files, and what additional data, validation, UX, engineering, governance, performance, or customer-readiness changes would be needed before presenting externally or scaling to other health systems. (2 paragraphs or less.)

Background & Data

Truveta works with health systems that want to understand clinical and operational performance, benchmark against peers or public references where appropriate, identify opportunities for quality improvement, and drive data-informed decisions. You are working with a simulated EHR-derived dataset from a hospital system – Cascade Regional Health. Your task is to create a minimum viable dashboard that would help a health system administrator understand important patterns in their patient population, encounters, diagnoses, and billed amounts.

Attached are four CSV files of simulated EHR data:

- patient.csv: patient demographics
- encounter.csv: encounter-level utilization

- billing.csv: encounter-associated billed amounts
- condition.csv: encounter-associated condition diagnoses using ICD codes

External benchmarks are optional. If used, they should be clearly sourced and accompanied by caveats about comparability to this simulated EHR-derived sample.

Objectives

We would like you to design and build an interactive dashboard for a health system administrator. The goal is to provide useful insight into the hospital system's clinical and operational performance and to show how the provided EHR data could support quality improvement, benchmarking, population health, utilization management, or similar health system priorities.

Build the dashboard using **Python Streamlit (preferred), R Shiny (preferred), Python Dash, or Python Shiny**; no other dashboarding frameworks should be used.

Your dashboard should help a user understand questions such as:

- What are the most important population, utilization, diagnosis, cost, or quality patterns in this hospital system?
- Where might there be opportunities for quality improvement, care management, operational improvement, or deeper investigation?

We do not expect perfect formal measure certification from this simulated dataset. We care more about your metric choices, assumptions, usability, and ability to tell a clear health-system story than about visual polish or application complexity.

You may use AI-assisted development tools for coding support, but your submitted work should reflect your own analytic judgment. Do not use generative AI tools, such as ChatGPT, to automatically identify data anomalies, generate dashboard insights, or create the analytic conclusions.

Please try not to spend more than 1-2 hours on this assignment. This is meant to be a sample of our approach to projects like this; we are not expecting perfection.

Data dictionary

Patient

- PatientId: Unique alphanumeric string identifying a patient.
- BirthDate: Patient birthdate.
- Sex: Sex of patient. Possible values include Male and Female. Values may be missing.
- Race: Race of patient. Possible values include Caucasian, African American, Asian, and Other.
- CensusRegion: Census region in which the patient resides. Possible values include Midwest, Northeast, South, and West.
- Zip3: First three digits of the patient's ZIP code

Encounter

- PatientId: Patient identifier.
- EncounterId: Unique alphanumeric string identifying an encounter.
- EncounterType: Type of encounter. Possible values include Outpatient and Inpatient.
- StartDate: Start date of the encounter.
- EndDate: End date of the encounter. Outpatient encounters may not have an end date.

Billing

- PatientId: Patient identifier.
- EncounterId: Encounter identifier.
- BillingDate: Date when the amount was billed.
- Amount: Amount billed for the encounter.

Condition

- PatientId: Patient identifier.
- EncounterId: Encounter identifier associated with the condition record.
- OnsetDate: Date when the condition was first recorded or observed for the associated encounter. For this simulated dataset, onset dates generally align with the associated encounter date range.
- Code: ICD diagnosis code.